Learning Program

King Mongkut's University of Technology Thonburi

Department of Mathematics, Faculty of Science

Course Title: MTH234 Linear Algebra **Section:** 1 (for CPE students in regular program)

Prerequisite: - **Semester:** 1/2025

Class Schedule: Tuesday at 10.30 am – 12.30 pm (Room CB2606) and 1.30 pm – 3.30 pm (Room 2506)

Lecturer: Dr. Anuwat Tangthanawatsakul **Office:** SCL801-15

Office Hours: Anytime by appointment **Email:** anuwat.sae@kmutt.ac.th

Course Description:

Finite dimension of vector spaces. Subspaces. Bases and dimension. Linear transformation. Matrices and linear operations. Determinants. Eigenvalues and eigenvectors. Diagonalization of matrices. Canonical forms for linear transformations. Quadratic forms.

Objectives:

CLO1 Students will be able to explain and prove main concept of vector space, dimension, basis and subspaces.

CLO2 Students will be able to compute matrix algebra, determinant, eigenvalues, and eigenvectors.

CLO3 Students will be able to explain and compute linear operations, linear operators and able to Interpret the results in applications.

CLO4 Students will be able to work effectively in team and have responsibility for assignment.

Teaching Methods: Give lecture by explanation and writing on paper (visualizer) or I-Pad (and possibly broadcast in ZOOM application)

Textbooks:

1. David Poole, Linear Algebra: A Modern Introduction, Third Edition, Cengage Learning, USA, 2015.
2. Ron Larson and David C. Falvo, Elementary Linear Algebra, sixth edition, Houghton Mifflin Harcourt Publishing Company, USA, 2009.

Course Assessment:

1. Exam 1 25%
2. Exam 2 25%
3. Final Exam 30%
4. Homework Assignments 12%
5. Project 8%

Teaching Program MTH 234

Week	Contents	Sub PLOs/CLOs/ Competence	Teaching Method / Assessment	Teacher
1	Matrices and Matrix Operations: Diagonal, Triangular, and Symmetric Matrices	CLO2	Lecture	Dr.Anuwat
2	Inverse of Matrices, Determinant, Elementary Row Operation	CLO 2	Lecture	Dr.Anuwat
3	Systems of Linear Equations	CLO 2	Lecture / Assignment	Dr.Anuwat
4	Vector Spaces, Subspaces	CLO 1	Lecture	Dr.Anuwat
5	Linear combination, and Span	CLO 1	Lecture	Dr.Anuwat
6	Exam 1: Wednesday 10th September 2025 at 9.00 – 11.00			
7	Linear Independence, Basis and Dimension	CLO 1	Lecture	Dr.Anuwat
8	Coordinates and change of basis	CLO 1	Lecture	Dr.Anuwat
9	Orthogonality and Gram-Schmidt Process	CLO 2	Lecture / Assignment	Dr.Anuwat
10	Linear Transformation	CLO 3	Lecture	Dr.Anuwat
11	Matrix Representation of Linear Transformation		Lecture	Dr.Anuwat
12	Exam 2: Wednesday 22nd October 2025 at 9.00 – 11.00			
13	Kernel, Range and Rank-Nullity Theorem	CLO 3	Lecture	Dr.Anuwat
14	Eigenvalues and Eigenvectors	CLO 2	Lecture	Dr.Anuwat
15	Diagonalization of matrices	CLO 2	Lecture	Dr.Anuwat
16	Canonical forms for linear transformations	CLO 2	Lecture / Assignment	Dr.Anuwat
17	Quadratic forms	CLO 2, CLO 3	Lecture	Dr.Anuwat
18	Final Exam: Wednesday 3rd December 2025 at 9.00 – 11.00			

Note: This schedule is flexible and is subject to change. Please attend the classes regularly.

Remark: Please join the course facebook group "**1_68 MTH234 Linear Algebra KMUTT (Section 1)**" by simply click the following link: https://www.facebook.com/groups/1695762188482157

The lecture notes, homework sets, and other class announcement will be regularly posted there.